



Photoabsorption and Photoionization in Gases

Daniel Vázquez Lago^{1,*}

¹ Department of Particle Physics, Cuadernos project, Santiago de Compostela, Spain

ARTICLE INFO

Keywords:
photoabsorption
photoionization
gas detectors
cross sections

ABSTRACT

A compact demonstration paper for testing the paper compilation pathway. We summarize the distinction between photon absorption and photoionization in a dilute gas and introduce the simplest attenuation model used to connect microscopic cross sections with macroscopic interaction probabilities.

1. Introduction

Photoabsorption denotes any process in which an incident photon transfers its energy to the medium, whereas photoionization is the subset of such interactions that leaves a free electron and a positive ion. In gaseous detectors the distinction matters because absorption controls photon transport, while ionization can seed additional charge and therefore modify the detector response.

2. Minimal interaction model

For a uniform gas with number density n , path length L , and an energy-dependent microscopic cross section $\sigma(E)$, a simple independent-interaction model gives

$$P(E, L) = 1 - \exp(-n\sigma(E)L). \quad (1)$$

Using $\sigma_{\text{abs}}(E)$ gives the total absorption probability, while $\sigma_{\text{ion}}(E)$ describes only the ionizing branch. Consequently, $\sigma_{\text{ion}}(E) \leq \sigma_{\text{abs}}(E)$ whenever non-ionizing absorption channels are available.

Table 1: Illustrative qualitative behaviour; entries are intentionally schematic.

Photon regime	σ_{abs}	σ_{ion}
Below threshold	finite or small	0
Near threshold	increasing	turn-on
Above threshold	channel dependent	finite

3. Photon transport context

In practical gaseous systems the relevant attenuation length depends strongly on photon energy, mixture composition, pressure, and the availability of ionizing and non-ionizing channels. The distinction becomes particularly useful when photons generated in an avalanche

can either disappear harmlessly or create a new electron that participates in subsequent multiplication.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod.

4. Discussion

The simple exponential model above neglects scattering, secondary emission, excited-state kinetics, and spatial variations of the gas. Nevertheless, it provides a useful baseline for checking units, threshold behaviour, and the relation between microscopic cross sections and detector-scale probabilities. A realistic implementation can replace the schematic cross sections by tabulated or calculated data without changing the structure of the transport model.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor.

5. Scope

This document is deliberately synthetic and exists to exercise the shared paper infrastructure, metadata parsing, style switching, two-column Elsevier layout, and output routing. Its scientific content can later be replaced by a full treatment without changing the compilation workflow.

* Corresponding author.

E-mail address: daniel@example.com (Daniel Vázquez Lago).

<https://doi.org/10.1016/j.mssp.2025.123456>

Received 23 June 2024; Received in revised form 23 June 2025; Accepted 10 December 2025

Available online 01 September 2025

0167-1234/© 2026 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleamur animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit.